

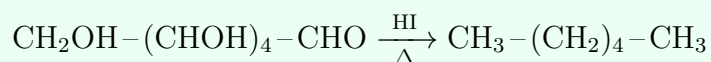
Chapter 10: Biomolecules

Reactions, Structures + Tips & Tricks

Formula Sheet

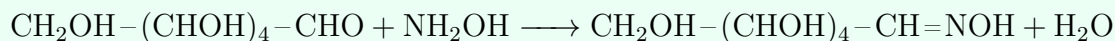
I. Carbohydrates – Structure-Establishing Reactions of Glucose

Reduction with HI (Confirms Straight Chain)



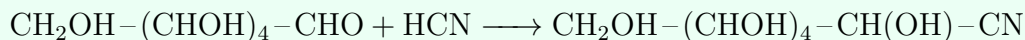
Glucose reduced all the way to n-hexane confirms all six carbons are in a straight chain.

Reaction with Hydroxylamine (Confirms -CHO Group)



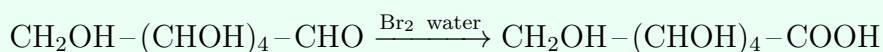
Forms an oxime, confirming the presence of a carbonyl (aldehyde) group.

Reaction with HCN (Confirms -CHO, Chain Extension)



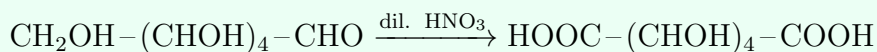
Forms a cyanohydrin; further confirms the carbonyl and is used in chain-extension proofs.

Bromine Water Test (Distinguishes Aldose from Ketose)



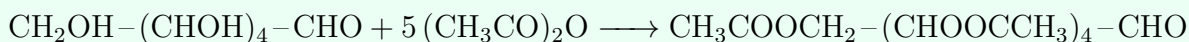
Glucose (aldose) is oxidised by mild bromine water to gluconic acid; fructose (ketose) is *not* oxidised by bromine water – this is the standard test to distinguish the two.

Reaction with Dilute HNO₃ (Confirms Symmetry / 1° and 6° positions)

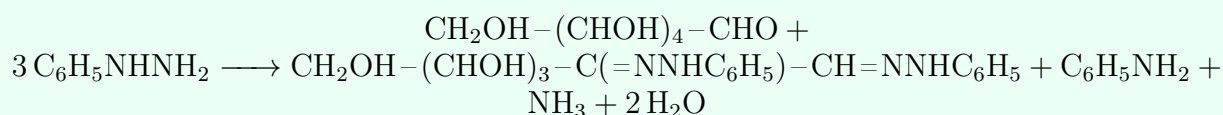


Both the -CHO and terminal -CH₂OH are oxidised to -COOH, forming saccharic acid; this dicarboxylic acid being optically inactive/meso-like helps establish the relative configuration.

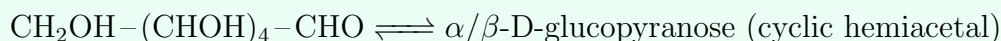
Acetylation (Confirms Five -OH Groups)



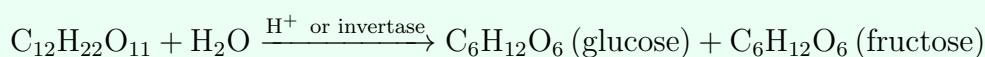
Forms glucose pentaacetate, confirming five -OH groups are present in the open-chain structure.

Osazone Formation (Phenylhydrazine, Confirms C1–C2)

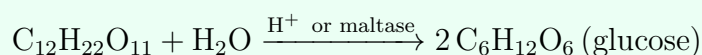
Excess phenylhydrazine reacts at C1 and C2 only, forming a bis-phenylhydrazone (osazone); glucose and fructose give the *same* osazone, proving they are identical from C3 to C6.

Cyclic Hemiacetal Formation

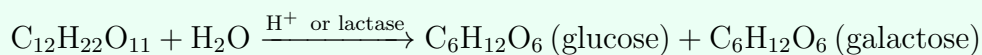
The -OH on C5 attacks the C1 carbonyl intramolecularly, forming a six-membered pyranose ring with a new -OH at C1 (anomeric carbon); explains mutarotation and why glucose does not give a Schiff's test.

II. Disaccharides – Glycosidic Linkage**Sucrose – Hydrolysis (Invert Sugar)**

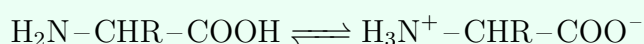
Sucrose is a non-reducing sugar: glycosidic linkage is between C1 of α -glucose and C2 of β -fructose, so both anomeric carbons are involved – no free -CHO/-C=O remains.

Maltose – Hydrolysis

Maltose is a reducing sugar: glycosidic linkage is α -1,4 between two glucose units, leaving one free anomeric -OH.

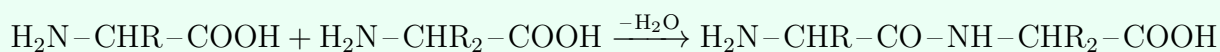
Lactose – Hydrolysis

Lactose (milk sugar) is a reducing sugar: β -1,4 glycosidic linkage between galactose and glucose, leaving one free anomeric -OH on the glucose unit.

III. Proteins – Amino Acids and Peptide Bond**Zwitterion Formation**

Amino acids exist predominantly as zwitterions (dipolar ions) in solution; the pH at which the molecule carries no net charge is the isoelectric point.

Peptide Bond Formation



Condensation between the $-\text{COOH}$ of one amino acid and the $-\text{NH}_2$ of another forms a peptide (amide) bond, releasing water; chains of many such bonds form polypeptides/proteins.

IV. Reference Tables

Classification of Carbohydrates

Monosaccharides – cannot be hydrolysed further; classified as aldoses (aldehyde group, e.g., glucose) or ketoses (ketone group, e.g., fructose).

Oligosaccharides – 2–10 monosaccharide units on hydrolysis; disaccharides (sucrose, maltose, lactose) are the most common.

Polysaccharides – many monosaccharide units; starch and glycogen are energy-storage polymers of α -glucose, cellulose is a structural polymer of β -glucose. Polysaccharides are *not* sweet and are non-reducing.

Reducing vs. non-reducing: a sugar with a free anomeric $-\text{OH}$ (free potential $-\text{CHO}/\text{C}=\text{O}$) is reducing (gives positive Fehling's/Tollens' test); one where both anomeric carbons are tied up in the glycosidic bond (e.g., sucrose) is non-reducing.

Protein Structure Levels

Primary – the linear sequence of amino acids joined by peptide bonds.

Secondary – regular folding patterns held by hydrogen bonds, e.g., α -helix and β -pleated sheet.

Tertiary – overall 3-D folding of a single polypeptide chain, giving the protein its specific biological activity.

Quaternary – spatial arrangement of two or more polypeptide subunits relative to each other (e.g., haemoglobin).

Denaturation – loss of secondary/tertiary/quaternary structure (biological activity lost) due to heat, pH change, etc., while the primary structure (peptide bonds) remains intact.

Vitamins – Classification and Deficiency

Fat-soluble: A (deficiency: night blindness/xerophthalmia), D (rickets), E (infertility), K (increased blood clotting time).

Water-soluble: B-complex and C – generally not stored in the body, so must be supplied regularly. B_1 deficiency: beriberi; B_2 : cheilosis; B_6 : convulsions; B_{12} : pernicious anaemia; C: scurvy.

Nucleic Acids – DNA vs. RNA

Basic unit is a **nucleotide** = nitrogenous base + pentose sugar + phosphate group. Bases are purines (adenine, guanine) and pyrimidines (cytosine, thymine/uracil).

DNA: sugar = deoxyribose; bases = A, T, G, C; double helical structure (Watson-Crick), strands held by H-bonding ($\text{A}=\text{T}$, $\text{G}\equiv\text{C}$); carries genetic information.

RNA: sugar = ribose; bases = A, U, G, C (uracil replaces thymine); usually single-stranded; helps in protein synthesis.

Tips, Tricks & Hacks

- **Glucose structure-proof chain, in order:** (1) HI reduction → straight 6-C chain; (2) reaction with HCN/NH₂OH → carbonyl present; (3) acetylation with 5 (CH₃CO)₂O → 5 -OH groups; (4) bromine water oxidises it (not fructose) → the carbonyl is specifically an aldehyde (C1); (5) HNO₃ oxidises both ends to -COOH giving an optically active saccharic acid → fixes the open-chain structure; (6) same osazone as fructose → C3–C6 are identical in both sugars.
- **Why glucose doesn't fully answer to Schiff's/2,4-DNP tests despite having -CHO:** in solution it predominantly exists as the cyclic hemiacetal (pyranose form), which has no free aldehyde – this is also why it shows mutarotation (changing optical rotation as α and β anomers interconvert through the open-chain form).
- **Reducing sugar rule of thumb:** if the sugar can revert to an open-chain form with a free -CHO or adjacent -C(OH)=, it's reducing. All monosaccharides are reducing; among common disaccharides, only sucrose is non-reducing.
- **Isoelectric point:** below it, the amino acid carries a net positive charge (protonated -NH₃⁺ dominates); above it, net negative (-COO⁻ dominates); exactly at it, the zwitterion has zero net charge and minimum solubility.
- **Denaturation vs. hydrolysis:** denaturation only disrupts secondary/tertiary/quaternary folding (physical), leaving peptide bonds (primary structure) intact; only chemical hydrolysis breaks the peptide bonds themselves.
- **DNA base-pairing memory aid:** "A-T, G-C" – Adenine pairs with Thymine (2 H-bonds), Guanine pairs with Cytosine (3 H-bonds) – G-C pairing is stronger.